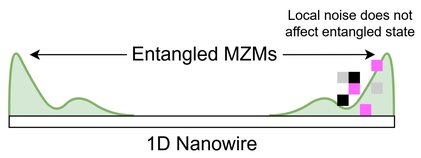
MEDA: Measurement-Efficient Disorder-Aware Majorana Zero Mode Detection in Realistic Devices

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Fault Tolerance via MZMs

Majorana Zero Modes (MZMs):
- non-Abelian quasiparticles
- emerge in topological superconductors
- Data stored in entanglement: **resilient to local noise**



Problem: How do we know MZMs exist?

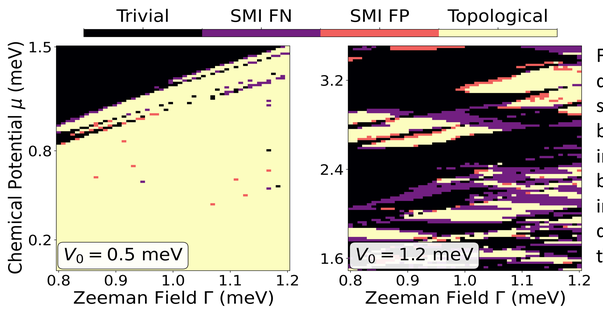
MZM Detection in Simulation

Prior **Topological Indicators** like the Scattering Matrix Invariant (SMI) attempt to translate rules from infinite, clean systems to realistic systems. Realistic systems have unique behaviors that bias topological indicators:

Finite size causes unexpected interaction between MZMs

Disorder fuzzes MZM-heralding features

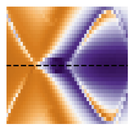
The **Periodic Disorder Invariant** (PDI) aims to eliminate such biases through disorder aware design. Instead of applying ideal rules to a finite disordered system, PDI considers the limit of replicating a finite disordered system.



For finite disordered systems, SMI becomes increasingly biased as disorder increases. PDI is designed to solve this problem.

MZM Detection in Real Systems

Topological indicators, including PDI, rely on unobservable information, making them infeasible for real systems. Machine learning approaches aim to connect observables to topological indicators but face severe measurement bottlenecks.

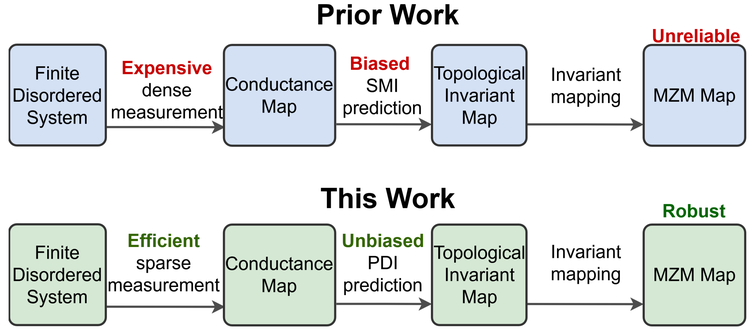


Each pixel in this conductance plot requires an independent measurement. Full parameter space coverage for one device requires 720,000 such images.

How can we integrate PDI with under a feasible measurement budget?

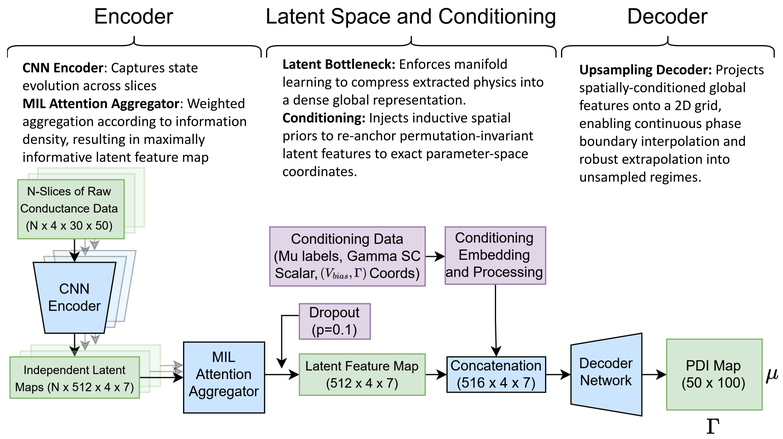
MEDA Overview

MEDA is a Measurement Efficient, Disorder-Aware ML framework for MZM detection in realistic devices.
- **Measurement Efficient**: MEDA uses a sparse parameter regime to decrease measurement requirements by 10x
- **Disorder-Aware**: MEDA maps observables to the PDI, which leverages system awareness to avoid biases



MEDA Architecture

MEDA utilizes an attention-driven multiple instance learning (MIL) architecture to identify the most topologically informative regions of the conductance map, enabling MEDA's sparse-measurement paradigm.



Experimental Efficiency: MEDA's architecture extracts global topological rules from sparse, unanchored conductance slices, interpolating complete and accurate phase diagrams without the need for exhaustive experimental sweeps.

Composite Loss Function

High disorder causes the topological phase space to drastically shrink and shatter, causing challenges for standard optimization strategies. To prevent the model from falling into a trivial-majority local minimum, MEDA utilizes a physically motivated composite loss function:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{BCE} + \lambda_2 \mathcal{L}_{Dice} + \lambda_3 \mathcal{L}_{Focal} + \lambda_4 \mathcal{L}_{TV}$$

Binary Cross Entropy: $\mathcal{L}_{BCE}$
Provides the foundational pixel-wise classification gradient for the model.

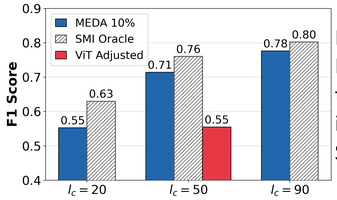
Focal Loss: $\mathcal{L}_{Focal}$
Dynamically down-weights the dominating trivial majority, increasing focus on rare, hard-to-predict pixels near quantum phase transitions.

Dice Loss: $\mathcal{L}_{Dice}$
Evaluates spatial overlap, heavily penalizing the model if it completely misses small, fragmented MZM islands.

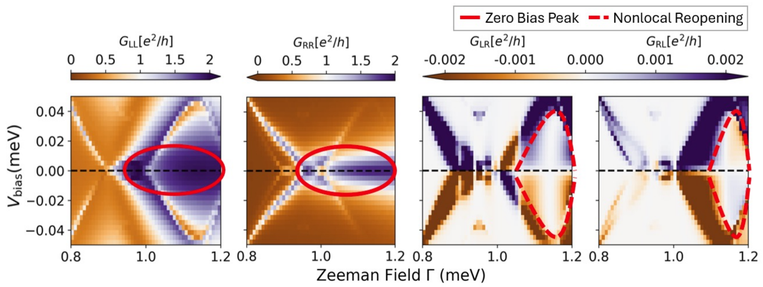
Total Variation Loss: $\mathcal{L}_{TV}$
Penalizes overfragmentation to prevent unrealistic noise artifacts common with sparse inputs. Recognized trade-off: limited ability to differentiate highly fragmented phase regions.

Scalar weights are optimized as hyperparameters to tune sensitivity and stability.

Results



Despite using only 10% of observable data, MEDA achieves performance comparable to the SMI Oracle and offers significant improvement over ViT, the state-of-the-art SMI-based prediction pipeline.



MEDA's attention weights heavily prioritize conductance maps with clearly visible conductance features that Morteza et al. establish as indicators of high topological phase probability.

Acknowledgments

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Full Paper